A CRITERION FOR RELATIVE VERY AMPLENESS IN TERMS OF IMMERSIONS INTO PROJECTIVE SPACE, FOR FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Six-line proof body carrying both directions of an equivalence, each direction a named reduction; the companion lemma for ample sheaves in the same section has the identical proof shape and is registered alongside this one (S1-morph-characterize-ample-ft).

Allowed standard prerequisites (editorial).

- The affine criterion for very ampleness of a finite type morphism. Lemma \ref{lemma-very-ample-finite-type-over-affine}, cited at proof lines 9603-9605. Gives the forward implication on each affine piece.
- Relative very ampleness for immersions into projective space. Lemma \ref{lemma-relatively-very-ample}, cited at proof lines 9606-9608. Gives the reverse implication.
- The hypothesis that the morphism is of finite type. the statement's assumption, used at proof lines 9602-9605. Makes the affine criterion applicable.

Source reference. Original lemma 02NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3489–3489. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02NP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3488–3488. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01VR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2456–2456. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible sheaf on X. Assume f is of finite type. The following are equivalent:*

(1) *$\mathcal{L}$ is f-relatively very ample, and*
(2) *there exist an open covering $S = \bigcup V_j$, for each j an integer n_j, and immersions*
$$i_j : X_j = f^{-1}(V_j) = V_j \times_S X \longrightarrow \mathbf{P}^{n_j}_{V_j}$$
over V_j such that $\mathcal{L}|_{X_j} \cong i_j^*\mathcal{O}_{\mathbf{P}^{n_j}_{V_j}}(1)$.

Proof. We see that (1) implies (2) by taking an affine open covering of S and applying Lemma 02NP to each of the restrictions of f and $\mathcal{L}$. We see that (2) implies (1) by Lemma 01VR. □